Chapter 7. AttRisk: An interface for predicting control arm attrition risk in newer glucose-lowering drug trials

7.1 Foreword

This chapter presents an overview and user guide for AttRisk, a proof-of-concept interactive Shiny application incorporating the framework developed in chapter 6. This chapter first introduces the purpose of AttRisk, the intended users, scope, and development process. The structure of the application is then described, including the main sections and distinctions between the trial design and trial conduct components. Guidance on how to use each component is then outlined, including required inputs, available outputs, interpretations of estimates, export options, and worked examples. Lastly, the chapter considers the strengths and limitations of AttRisk, and opportunities to improve and extend the scope and functionality.

7.2 Introduction

AttRisk is an interactive Shiny application that estimates attrition risk for control arms in trials of newer glucose-lowering drugs for type 2 diabetes mellitus. The application was developed using the design-stage and conduct-stage risk prediction models described in chapter 6. The trial design component estimates attrition risk using trial design features, whereas the trial conduct component estimates participant risk profiles within a specified trial design context by combining design-stage predictions with participant baseline characteristics.

AttRisk was developed to demonstrate how risk predictions could design choices, retention planning, and risk-based monitoring during follow-up. The application is designed for users who are interested in understanding how attrition might occur over time for a specific trial design scenario, and how attrition risk may vary between participant profiles within a given trial context. Intended users include trialists, clinicians, methodological researchers, and other members of trial teams involved in planning or evaluating retention.

The current version of AttRisk applies to control arms in phase III/IV trials of newer glucose-lowering drugs for type 2 diabetes mellitus. The decision to develop risk prediction models for control arms was described in chapter 6 and was informed by the findings of chapter 5, where attrition was found to be higher among control arms. In this guide, control arms refer to placebo and/or active comparator arms, such as older glucose-lowering drugs. Estimates produced by AttRisk should therefore be interpreted within this scope. AttRisk is not intended to determine participant eligibility, motivate the exclusion of participants from trials, replace clinical judgement, or predict individual participant behaviour with a high degree of certainty. The application should be viewed as a scenario-building and decision-support tool that generates outputs to inform retention planning and monitoring, and to identify possible retention risks.

7.3 AttRisk development

AttRisk was developed using a standard Shiny framework that separated the user interface, server-side functionality, and global resources into standalone scripts. The interface was designed as a dashboard with sidebar navigation, allowing a user to move between a home page, user guide, trial design component, trial conduct component, and an about page.

Several usability features were incorporated to guide application use and reduce input errors. Action buttons in the trial design and trial conduct components are disabled by default until the required inputs have been provided. Reset buttons allow users to clear some or all of the inputs, depending on the component being used. Numeric input validation rules were also incorporated to prevent users from generating predictions using values outside the permitted ranges.

AttRisk uses the prediction models described in chapter 6, but users do not need to interact with the statistical model directly. In the design-stage component, the app takes in information about the planned trial to estimate how attrition risk may build up over time. In the conduct-stage component, the app starts with the trial design scenario and adjusts predicted risk according to the baseline characteristics of a hypothetical participant profile. The app applies the log-rate adjustment described in chapter 6 when combining these information sources. All of these steps are handled automatically within the app, allowing

users to work with familiar trial and participant inputs rather than statistical model parameters.

The full application framework is available at the PhD GitHub repository (github.com/RMcCPhD/attrisk_shiny). AttRisk was developed using R 4.5.1 (R Foundation for Statistical Computing; Vienna, Austria). The core packages used were *shiny* and *shinydashboard*. Additional packages included *shinyjs* for dynamic control of interface behaviour, *shinycssloaders* to incorporate loading animations while outputs were generated, *shinyalert* to create an introductory disclaimer, *shinyvalidate* to validate user inputs, *tidyverse*, *DT*, and *kableExtra* for data transformations and output generation, and *markdown* and *knitr* to create the in-app user guide.

7.4 Application overview

AttRisk is an interactive dashboard, with navigation provided through a sidebar menu. This menu allows users to move between the home page, user guide, trial design component, trial conduct component, and about page. The home page provides a brief introduction to the application and orientates the user. The user guide links to this document. The trial design and trial conduct components provide the core functionality of AttRisk. The about page provides additional information about the application, authors and collaborators, opportunities to provide feedback, and the application version.

7.4.1 AttRisk estimands

In AttRisk, attrition is defined as participant non-completion of a trial for any reason by the end of the user-specified follow-up duration. The primary estimand is attrition risk over time, defined as the cumulative probability of attrition from randomisation to a user-specified time during follow-up. For example, an estimated attrition risk of 21% by week 26 would correspond to approximately 210 non-completers among 1,000 participants randomised to control arms by that time.

In addition to attrition risk, the design component displays attrition hazard, defined as the instantaneous rate of attrition at a given time mong participants who remain in follow-up. Hazard is displayed on a relative scale describing when attrition is expected to occur most frequently. This scaling aids interpretation, with values falling between 0, when attrition is least frequent, and 1, when

attrition is most frequent. Scaled values should not be interpreted as probabilities and cannot be compared to risk estimates.

7.4.2 Interpreting AttRisk predictions

AttRisk estimates are model-based predictions intended to support a broad interpretation of attrition risk patterns, comparisons between plausible trial scenarios, and to convey the relationship between trial design features, participant profiles, follow-up, and attrition. Users should avoid treating estimates as exact forecasts for existing or future trials. The visual representations of uncertainty in plots describe uncertainty under the fitted model, but cannot currently establish the accuracy of predictions in a new trial. Chapter 6 internally validated the design-stage model, but the conduct-stage component was not internally validated, and neither component has been externally validated.

To reinforce this limitation, AttRisk displays a disclaimer when first opened: *'AttRisk is a prototype application built on models that were developed as part of a PhD project. Predictive capabilities have not been clinically validated and may not reflect real-world outcomes. Please use with appropriate caution.'* (Figure 7.1).

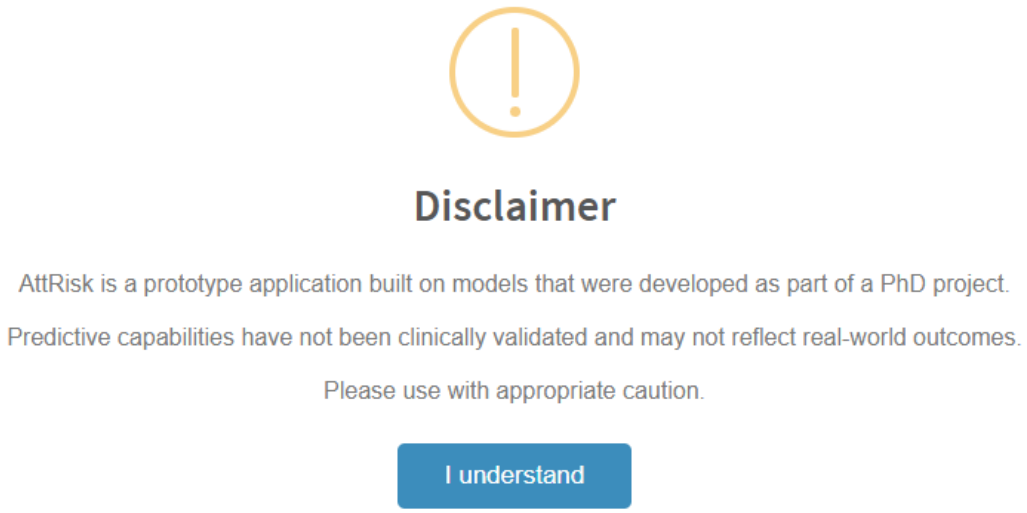


Figure 0.1 Disclaimer displayed when opening AttRisk for the first time

The trial conduct component includes participant characteristics such as age, sex, race/ethnicity, and markers of general and metabolic health. Differences in predicted risk between profiles defined by these characteristics should not be interpreted as inherent participant deficits. Instead, they may indicate broader

structural, social, cultural, healthcare-related, or operational barriers to retention. Participant-level attrition risk should therefore be interpreted as a signal for retention planning and further evaluation, not as a reason to avoid participant enrolment.

Some user inputs are marked with an asterisk within the trial design and trial conduct components. This indicates design features and characteristics that were identified as credible predictors of attrition risk in the corresponding models. This notation is intended to highlight inputs that demonstrated credible associations with attrition risk, but does not imply that these inputs are causal in a broader clinical, social, or operational context.

7.4.3 Common interface behaviours

In both the trial design and trial conduct components, users follow a structured sequence of input specification, prediction generation, output interpretation, and optional export. Action buttons that initiate risk estimation and generate outputs remain disabled until all required inputs are entered. If a numeric value falls outside of a permitted range, the user will be prompted to correct the value before predictions can be generated. Once all required inputs are entered and valid, action buttons become active and users can generate predictions, reset or clear inputs and outputs, and save outputs.

7.5 Trial design

7.5.1 Purpose

The trial design component of AttRisk allows a user to estimate attrition risk, based on trial design features, for the control arm(s) of a hypothetical trial of newer glucose-lowering drugs for type 2 diabetes mellitus. This component is intended for use during trial planning when a trial team may wish to understand how predicted attrition may differ across trial design scenarios. Figure 7.2 presents an example of the trial design component in use.

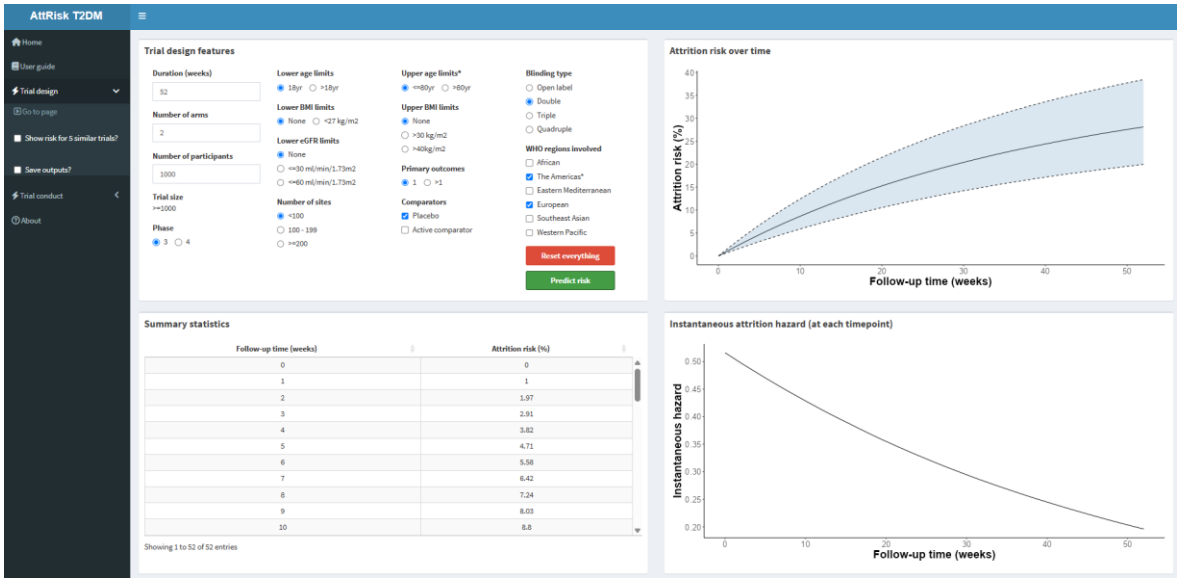


Figure 0.2 The trial design component of AttRisk in use

7.5.2 Interface

The trial design component is organised into one input box and three output boxes. In this section, the user inputs are referred to as Box A, the attrition risk over time plot as Box B, the summary statistics as Box C, and the attrition hazard plot as Box D.

The user inputs (Box A), presented in Figure 7.3, contain the trial design features available to the user. The full set of trial design features, including value ranges, requirements, and reference values, is presented in Table 7.1. The input options correspond to the design-stage model covariates and reflect routinely reported trial design features. Users enter the required inputs and can retain or change the reference values for the remaining trial design features.

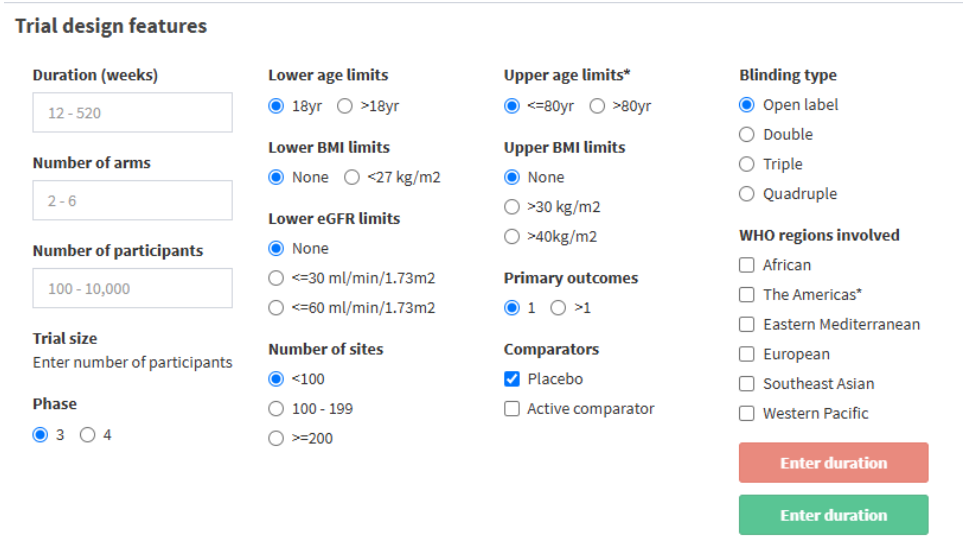


Figure 0.3 Box A: Available inputs for trial design features

Table 0.1 Trial design features available as user inputs

Input	Description	Values/Range	Default value	User input required
Duration	Follow-up in weeks	12 to 520	None	Yes
Number of arms	Number of treatment arms	2 to 6	None	Yes
Number of participants	Total randomised participants	100 to 10,000	None	Yes
Trial size	Categorised number of participants	Small (<500); Medium (500-999); Large (≥1000)	Calculated automatically	No
Upper age limits	Restrictions for maximum participant age	70 years; >70 years; None	None	No
Lower age limits	Restrictions for minimum participant age	18 years; >18 years; None	None	No
Upper body mass index (BMI) limits	Restrictions for maximum participant BMI	<45 kg/m^2; None	None	No
Lower BMI limits	Restrictions for minimum participant BMI	≤25 kg/m^2; None	None	No
Lower estimated glomerular filtration	Restrictions for minimum participant	≥60mL/min/1.73m^2; ≥30mL/min/1.73m^2; None	None	No

rate (eGFR) limits	kidney function			
Phase	Trial phase	3; 4	3	No
Number of POMs	Number of primary outcomes	1; >1	1	No
Comparators	Treatment comparator(s)	Placebo; active comparator; both	Placebo	No
Blinding	Blinding type	Open label; double; triple; quadruple	Open label	No
WHO regions	WHO regions involved	African; the Americas; Eastern Mediterranean; Europe; South-east Asia; Western Pacific	None	No

Box B (Figure 7.4) displays a plot of attrition risk over time. The x-axis represents follow-up duration in weeks, from baseline (week 0) to the end of follow-up. The y-axis represents predicted cumulative attrition risk expressed as a percentage. The black line represents the central predicted attrition risk, the blue shading represents uncertainty intervals, and the dashed black lines indicate the upper and lower interval limits. Wider intervals indicate greater uncertainty in the predicted attrition risk.

Box C (Figure 7.5) displays summary statistics for predicted attrition risk over time, showing cumulative attrition risk at each week of follow-up.

Box D (Figure 7.6) displays the attrition hazard. The x-axis represents the follow-up duration in weeks, from baseline to the end of follow-up. The y-axis displays hazard on a relative scale as described in section 7.4.1. The plot indicates how the rate of attrition among participants remaining in follow-up changes over time.

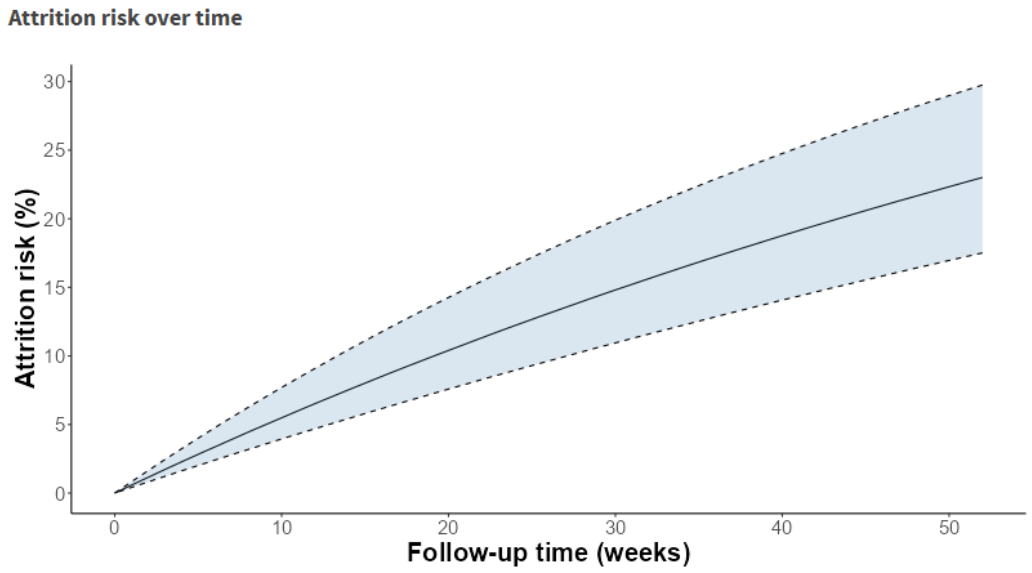


Figure 0.4 Box B: Output plot illustrating attrition risk over time for the trial design component

Summary statistics

Follow-up time (weeks)	Attrition risk (%)
0	0
1	1
2	1.97
3	2.91
4	3.82
5	4.71
6	5.58
7	6.42
8	7.24
9	8.03
10	8.8

Showing 1 to 52 of 52 entries

Figure 0.5 Box C: Output summary of attrition risk over time for the trial design component

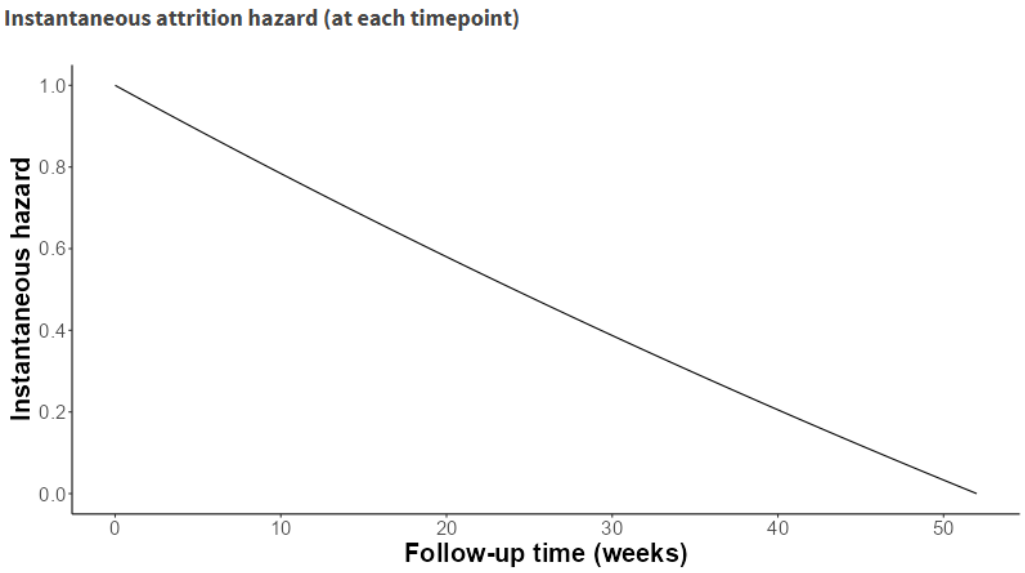


Figure 0.6 Box D: Output plot illustrating attrition hazard for the trial design component

7.5.3 Inputs and validation

Three trial design inputs are required before predictions can be generated: trial duration, number of arms, and planned sample size. The action buttons, namely ‘Predict risk’ and ‘Reset everything’, remain disabled until these inputs have been entered and validated. While required inputs are missing, the action button labels indicate the next required input, displaying either ‘Enter duration’, ‘Enter arms’, or ‘Enter sample size’ (Figure 7.7).

The permitted values for required inputs are 12 to 520 weeks for trial duration, 2 to 6 for the number of arms, and 100 to 10,000 for the number of participants. If a value outside the permitted range is entered, the interface displays a warning prompting the user to correct the value before continuing (Figure 7.8). When the number of participants is entered, trial size will be automatically categorised as either small (<500 participants), medium (500-999 participants), or large (≥1000 participants). This category is displayed below the number of participants. All other design features are set at default/reference values, which users can retain or modify to reflect a hypothetical trial scenario.

These limits are interface checks and not evidence that every permitted scenario is supported by the development data. Users should review whether the selected design features form a plausible trial scenario.

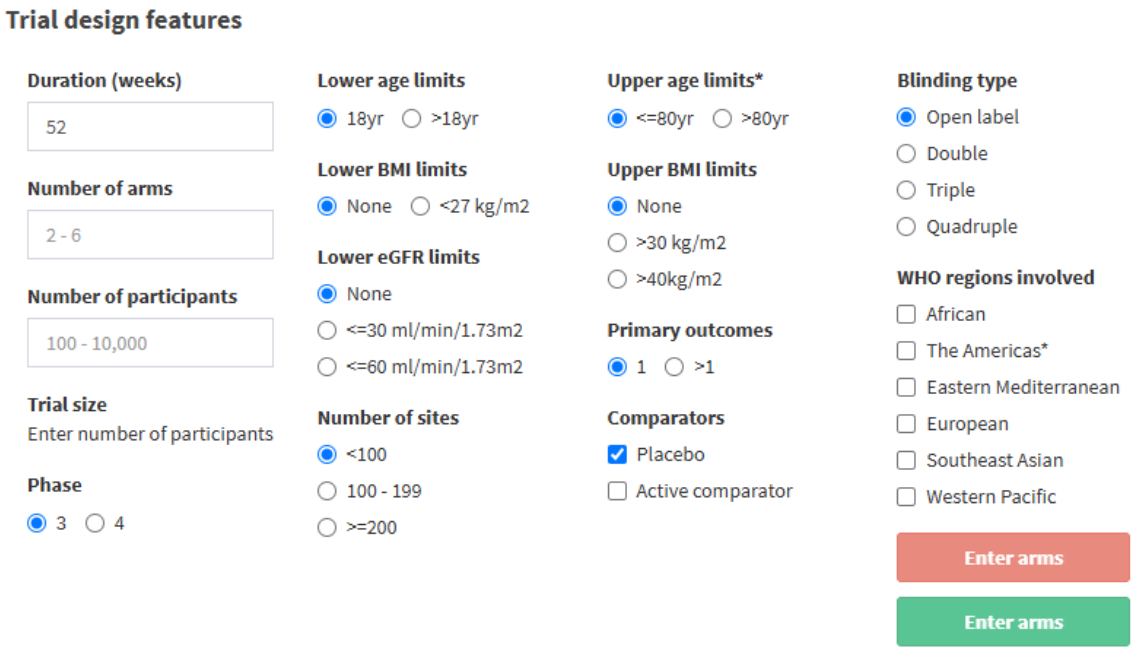


Figure 0.7 Example of disabled action buttons and prompts for the required input

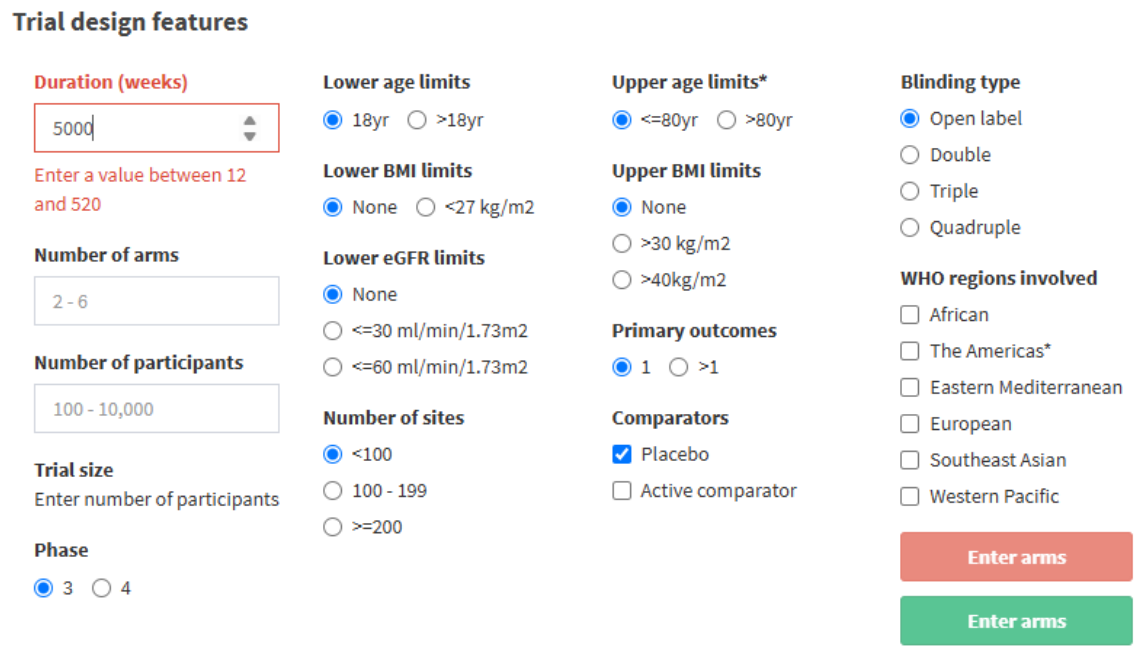


Figure 0.8 Example warning message prompting a user to enter a plausible input value

7.5.4 Generating predictions

To generate predictions for a hypothetical trial, the user:

1. Opens the trial design page from the sidebar menu
2. Enters trial duration in weeks
3. Enters the number of arms
4. Enters the number of participants
5. Reviews or modifies the remaining trial design features
6. Clicks ‘Predict risk’

After ‘Predict risk’ is clicked, loading indicators appear while the outputs are generated. Once generated, Boxes B, C, and D are populated with the corresponding outputs. Users can then reset the interface and create a different trial scenario.

7.5.5 Comparing outputs to existing trials

The trial design sidebar menu includes an optional checkbox labelled ‘Show risk for 5 similar trials?’. This checkbox can be selected after the design outputs are generated. When selected, AttRisk identifies five trials in the model development dataset with design features most similar to the user-specified scenario. A risk summary is displayed alongside the five trial identifiers, and each identifier links to the corresponding ClinicalTrials.gov page (Figure 7.9).

Markers are added to Box B to provide descriptive benchmarks for predicted attrition risk (Figure 7.10).

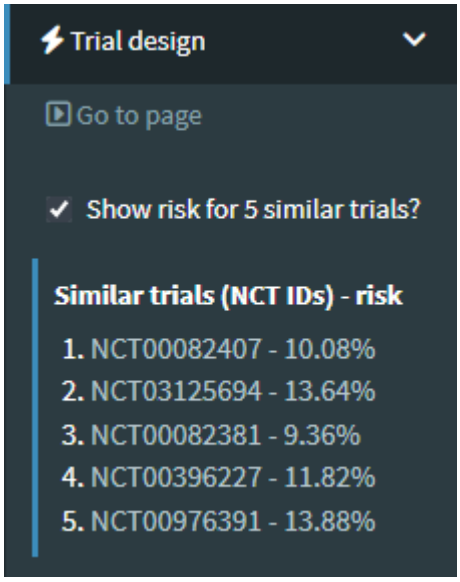


Figure 0.9 Example of output from selecting 'Show risk for 5 similar trials?'

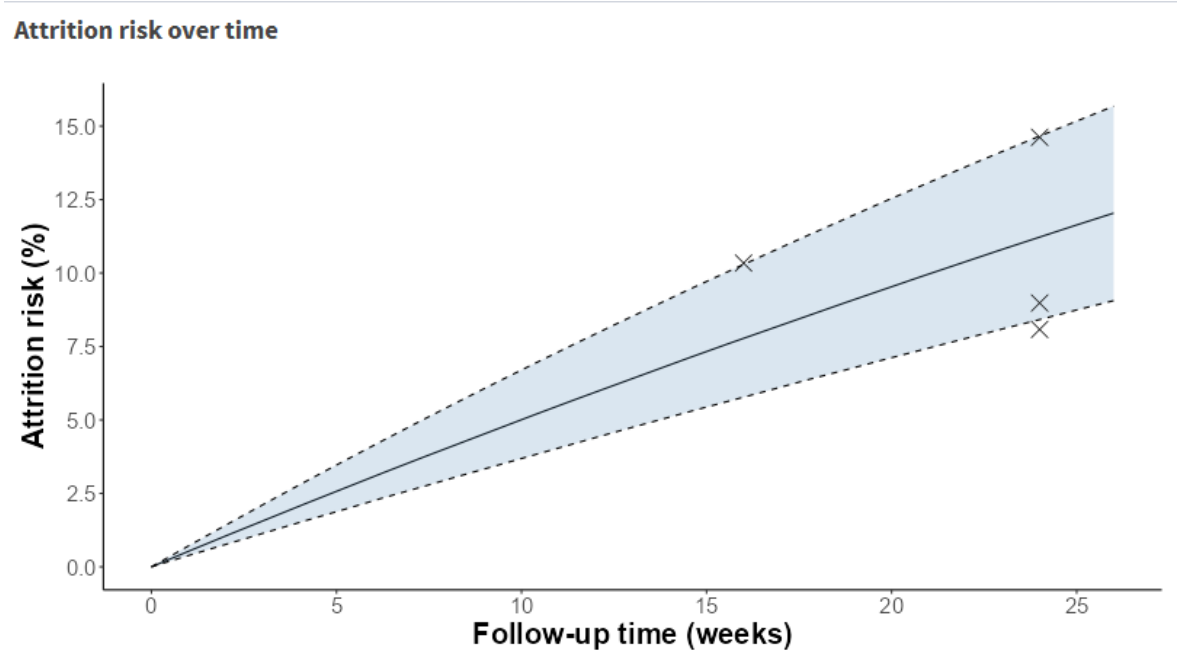


Figure 0.10 Markers added to Box B plot indicating attrition risk in five similar trials

7.5.6 Saving outputs

Trial design outputs can be exported by selecting the 'Save outputs?' checkbox in the trial design sidebar menu. When selected, an additional set of checkboxes appears, allowing the user to choose which outputs to save (Figure 7.11).

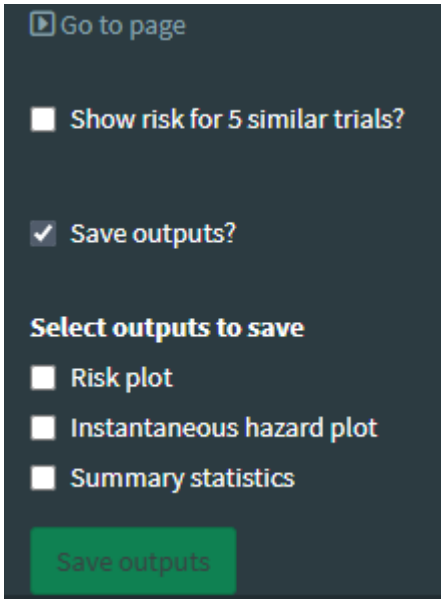


Figure 0.11 Example of the save options presented after selecting 'Save outputs?'

In addition to any selected outputs, a CSV file containing the trial design features selected by the user is saved. If the user selected the option to display five trials with similar design features, the export also includes information about these trials. After selecting the desired outputs, the user clicks 'Save outputs' to initiate the process. A loading animation indicates that the files are being prepared. Once prepared, a popup appears indicating the files are ready to be saved (Figure 7.12). The user can then select 'Download zip' to export a zipped file containing the outputs to their local machine. The user can then close the popup and return to the interface.

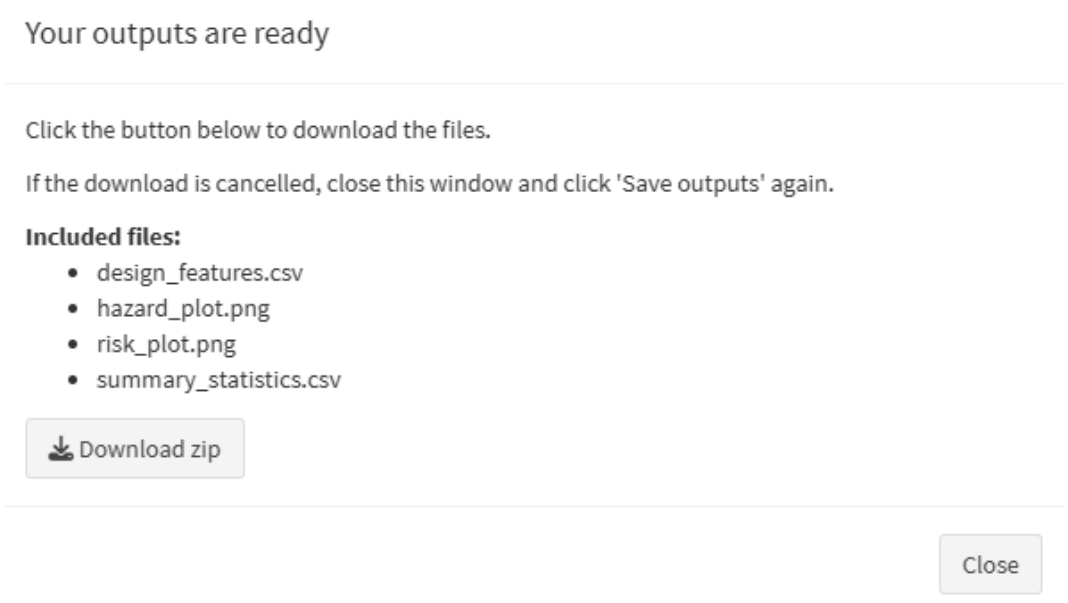


Figure 0.12 Popup window displayed allowing the user to export the desired outputs

7.5.7 Example: Comparing trial design scenarios

A trial team is designing a phase III randomised controlled trial evaluating a newer glucose-lowering drug for adults with type 2 diabetes mellitus. The team expects the trial will include three arms: two experimental treatment arms and one non-experimental comparator arm. Before finalising the protocol, the team use AttRisk to understand how attrition might occur under different design assumptions, and whether retention planning should focus on particular periods of follow-up.

The team first consider a 26-week, phase 3, double-blinded trial with three arms, a placebo comparator, 250 randomised participants, and sites in the Americas and Europe. The team enter these design features into AttRisk and retain the default/reference values for the remaining trial design features, including eligibility criteria, number of sites, and number of primary outcomes. The interface after generating outputs is presented in Figure 7.13.

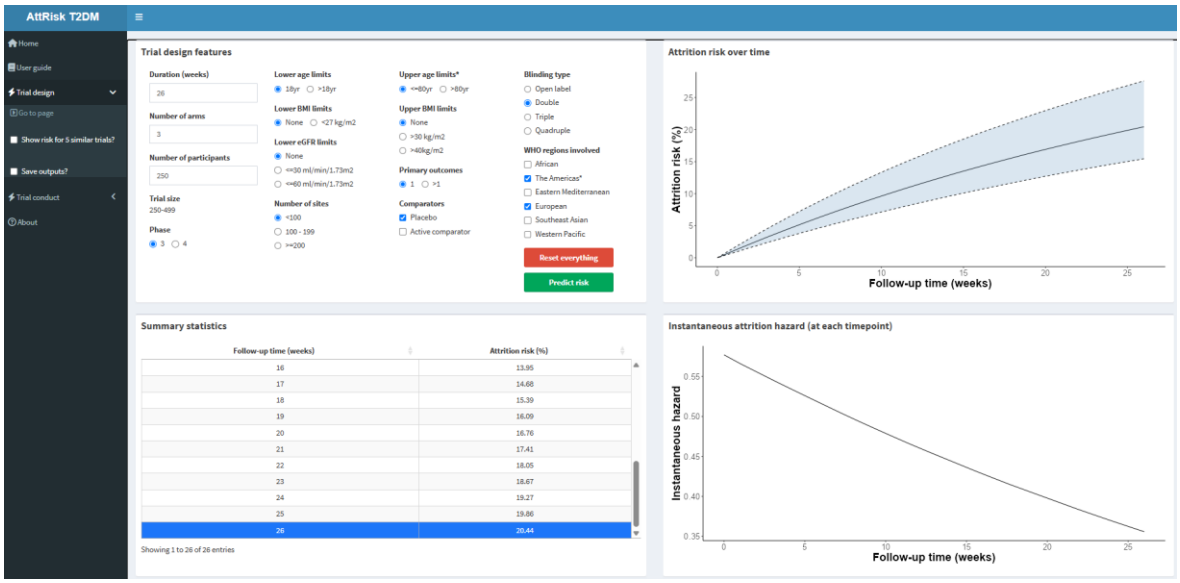


Figure 0.13 Outputs generated in the first example scenario

The attrition risk plot shows how attrition may accumulate over the 26-week follow-up period, and the summaries indicate that attrition risk may be approximately 20% by the end of follow-up. The hazard plot indicates that the attrition rate among participants remaining in follow-up may be higher toward the beginning of follow-up. The team repeat this exercise with a larger sample size of 500 randomised participants and keep all other features the same (Figure 7.14). Compared with the previous scenario, the team find that predicted attrition risk may be similar by the end of follow-up, at approximately 19%.

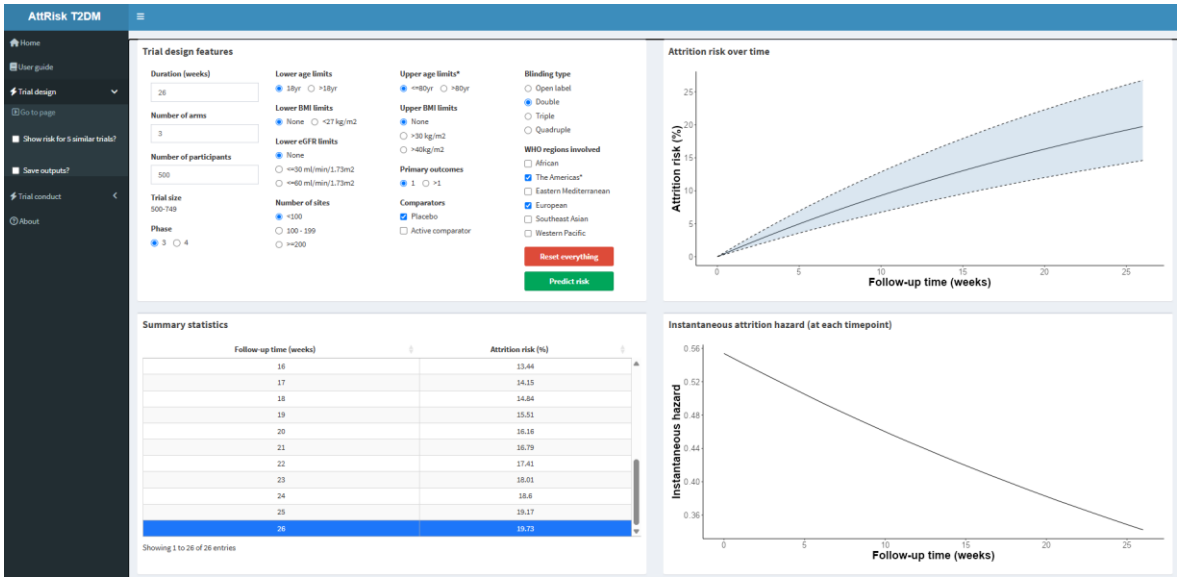


Figure 0.14 Outputs generated in the second example scenario

The difference between these scenarios is small, suggesting that increasing the sample size by 250 participants would have a limited impact on attrition. For retention planning, the team could use the early hazard pattern to inform when additional contact or review of follow-up difficulties might be considered.

7.6 Trial conduct

7.6.1 Purpose

The trial conduct component of AttRisk allows a user to estimate attrition risk for a hypothetical participant profile within a specified trial design context. This extends the trial design component by combining trial design features with participant baseline characteristics corresponding to the conduct-stage risk prediction model covariates. This component is intended for scenarios where users wish to explore how attrition risk may vary between participant profiles randomised to a control arm in a trial of newer glucose-lowering drugs for type 2 diabetes mellitus. Figure 7.15 presents an example of the trial conduct component in use.

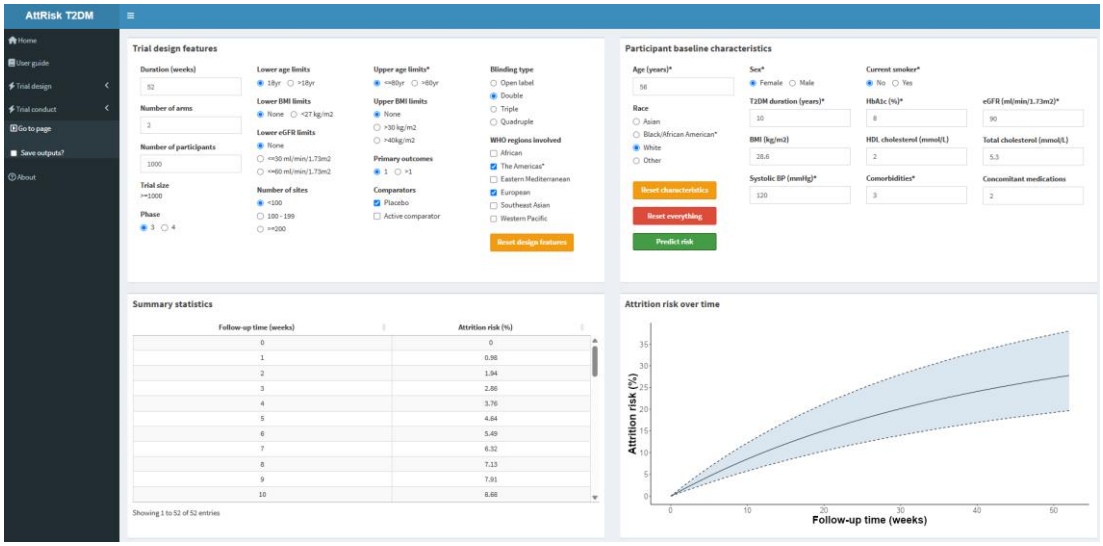


Figure 0.15 The trial conduct component of AttRisk in use

7.6.2 Interface

The trial conduct component is organised into two input boxes and two output boxes. The trial design inputs are referred to as Box A, the participant baseline characteristics as Box B, the summary statistics as Box C, and attrition risk over time plot as Box D.

Box A contains the trial design features, corresponding to the descriptions provided in section 7.5. Box B (Figure 7.16) contains the participant baseline characteristics used to construct the participant risk profile. Available baseline participant characteristics are outlined in Table 7.2 and correspond to the covariates included in the conduct-stage model. Box C and Box D contain the summary statistics and plot of attrition risk over time, respectively, corresponding to the output formats described in section 7.5.

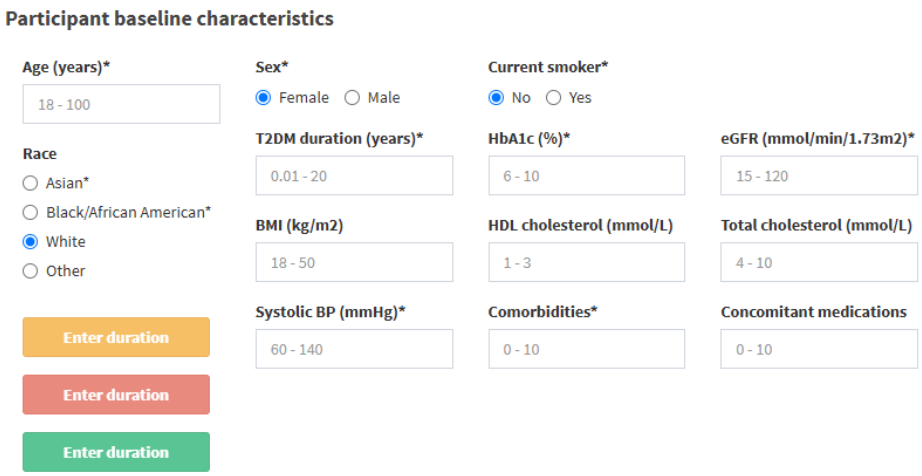


Figure 0.16 Box B: Available inputs for baseline participant characteristics

Table 0.2 Participant baseline characteristics available as user inputs

Input	Description	Values/Range	Default value	User input required
Age	Age in years	18 to 100	None	Yes
Sex	Participant sex	Female; Male	Female	No
Current smoker	Smoking status	No; Yes	No	No
Race/Ethnicity	Categorised race/ethnicity	Asian; Black/African American; White; Other	White	No
Duration of type 2 diabetes mellitus (T2DM)	Time since T2DM diagnosis in years	0.01 to 20	None	No
Hb_{A1c}	Glycaemia in %/DCCT units	6 to 10	None	No
Estimated glomerular filtration rate (eGFR)	Kidney function in $mL/min/1.73m^2$	15 to 120	None	No
Body mass index (BMI)	BMI in kg/m^2	18 to 50	None	No
High density lipoprotein (HDL) cholesterol	HDL cholesterol in mmol/L	1 to 4	None	No
Total cholesterol	Total cholesterol in mmol/L	4 to 10	None	No
Systolic blood pressure (BP)	Systolic BP in mmHg	90 to 200	None	No
Number of comorbidities	Count of comorbid conditions	0 to 10	None	No
Number of concomitant medications	Count excluding antidiabetics	0 to 20	None	No

7.6.3 Inputs and validation

Four inputs are required to generate trial conduct predictions: trial duration, number of arms, number of participants, and participant age. The trial conduct action buttons, namely ‘Predict risk’, ‘Reset design features’, ‘Reset characteristics’, and ‘Reset everything’, remain disabled until these inputs have been entered and validated. While required inputs are missing, action button text will indicate the required inputs as either ‘Enter duration’, ‘Enter arms’, ‘Enter sample size’, or ‘Enter age’(Figure 7.17).

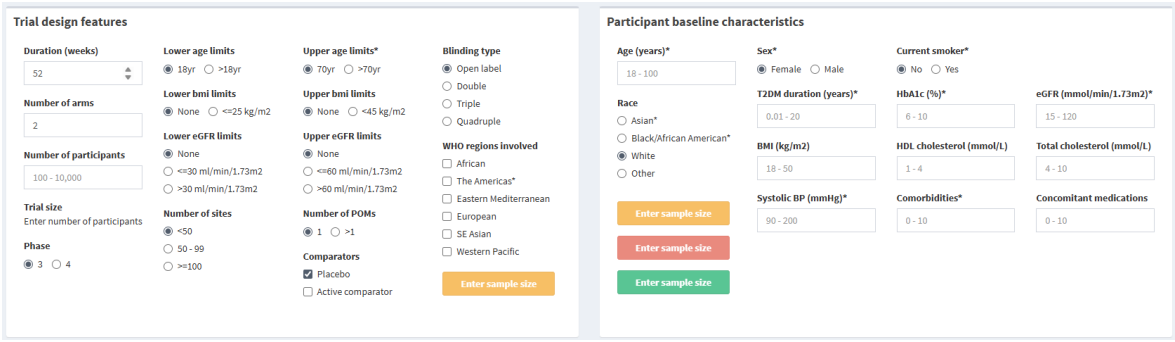


Figure 0.17 Example of disabled action buttons and prompts for the required inputs

The permitted values for trial design features are the same as those described in section 7.5.3. For participant baseline characteristics, permitted values are 18 to 100 years for age, 0.01 to 20 years for type 2 diabetes mellitus duration, 6 to 10% for Hb_{A1c}, 15 to 120 mL/min/1.73m^2 for eGFR, 18 to 50 kg/m^2 for BMI, 1 to 4 mmol/L for HDL cholesterol, 4 to 10 mmol/L for total cholesterol, 90 to 200 mmHg for systolic blood pressure, and 0 to 10 for number of comorbidities, and 0 to 20 for concomitant medications. If a value outside of a permitted range is entered, a warning structured similarly to that described in section 7.5.3 prompts the user to enter a plausible value. Once all required inputs are entered and are valid, the action buttons activate and the user can generate predictions.

Where optional continuous characteristics are left blank, AttRisk assigns values of zero, corresponding to the pooled mean used during model development. This contributes no additional change to participant-level linear predictors. To make comparisons reproducible, it is recommended that users save outputs which includes entered values (see section 7.6.5).

7.6.4 Generating predictions

To generate predictions for a hypothetical participant profile, the user:

1. Opens the trial conduct page from the sidebar menu
2. Enters trial duration in weeks
3. Enters the number of arms
4. Enters the number of participants
5. Reviews or modifies the remaining trial design features
6. Enters participant age
7. Reviews or modifies the remaining participant baseline characteristics
8. Clicks ‘Predict risk’

After clicking ‘Predict risk’, loading indicators appear while outputs are generated. Boxes C and D are then populated with the summary statistics and attrition risk plot. The user can then modify participant characteristics and/or trial design features to generate a different participant profile within the same or a different trial design context. The user can reset either set of inputs by clicking ‘Reset design features’ or ‘Reset characteristics’. To completely reset all inputs and outputs, the user can select ‘Reset everything’.

7.6.5 Saving outputs

The user can save outputs from the trial conduct component in the same manner as described in section 7.5.6. When saving outputs, the exported CSV file contains both the inputted trial design features and baseline participant characteristics.

7.6.6 Example: Comparing participant risk profiles

Following the example in section 7.5.7, the trial team has finalised the design of the trial evaluating a newer glucose-lowering drug in adults with type 2 diabetes mellitus. The agreed trial design features are: double-blinding, three arms (two experimental, one placebo), 500 randomised participants, recruitment in the Americas and Europe across fewer than 50 sites, 26 weeks of follow-up, no restrictions on age, body mass index, or kidney function, and one primary outcome (change from baseline Hb_{A1c}). Having used the trial design component of AttRisk to inform planning, the team use the trial conduct component to compare attrition risk trajectories across hypothetical participant profiles (Table 7.3). These profiles are hypothetical, and any alignment with real individuals,

either externally or within the IPD used to develop the conduct-stage models, is coincidental.

Table 0.3 Participant profiles generated from baseline participant characteristics

Baseline characteristics	**Participant 1**	**Participant 2**	**Participant 3**
Age, years	45	60	35
Sex	Male	Male	Female
Race/Ethnicity	White	Black/African American	Asian
Current smoker	No	Yes	No
T2DM duration, years	2	18	3
BMI, kg/m^2	28.9	32.3	41
Hb_{A1c}, %	8	8.5	6.3
eGFR, $mL/min/1.73m^2$	70	40	100
Comorbidities	2	3	0
Concomitant medications	2	2	2

The outputs generated for each participant profile are presented in Figures 7.18-7.20 and summarised in Table 7.4. Across profiles, the team find that participant 3 may have the lowest attrition risk by the end of follow-up, while participant 2 may have the highest attrition risk.

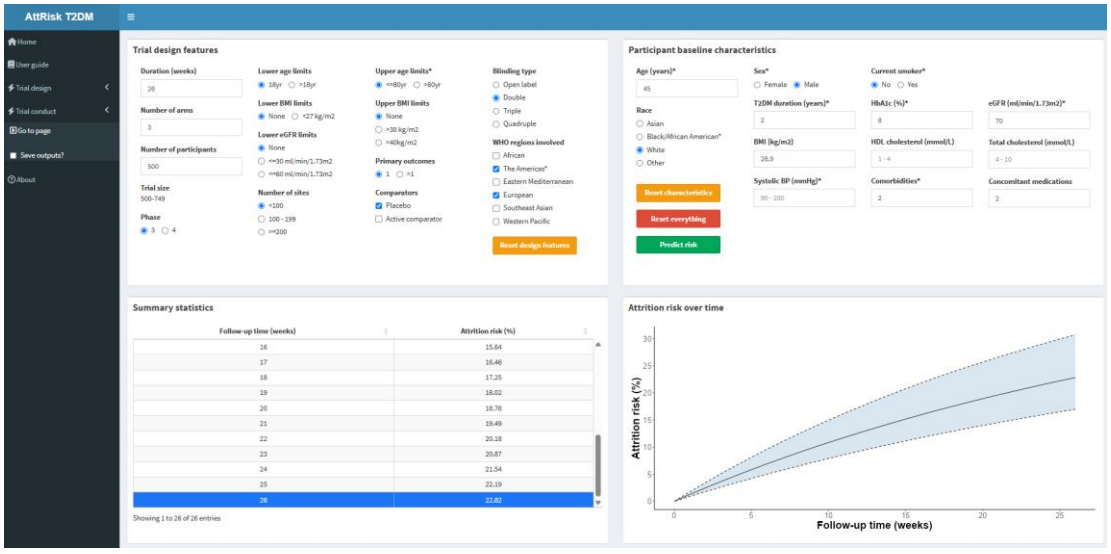


Figure 0.18 Outputs generated for participant 1 profile

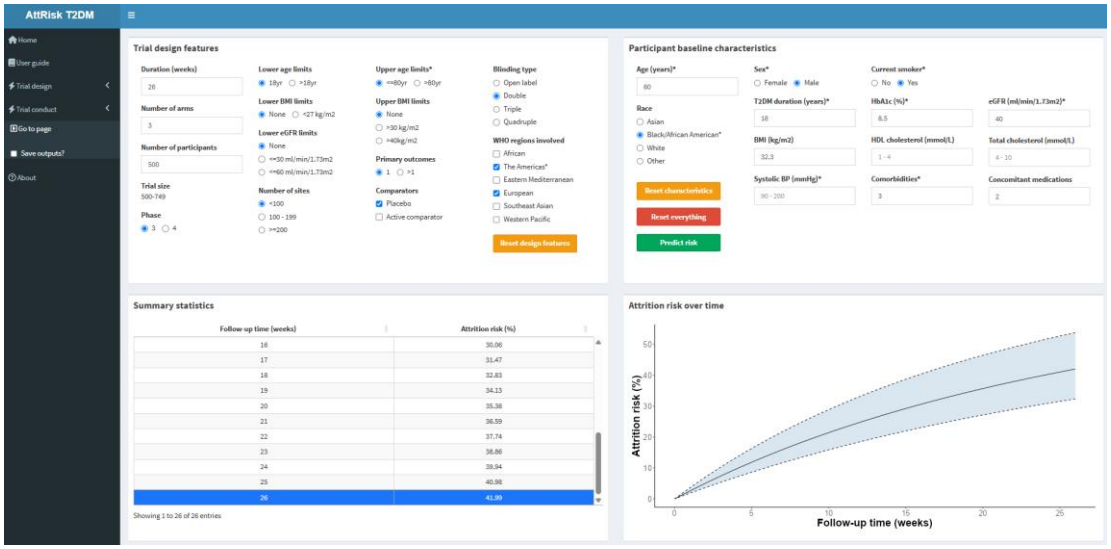


Figure 0.19 Outputs generated for participant 2 profile

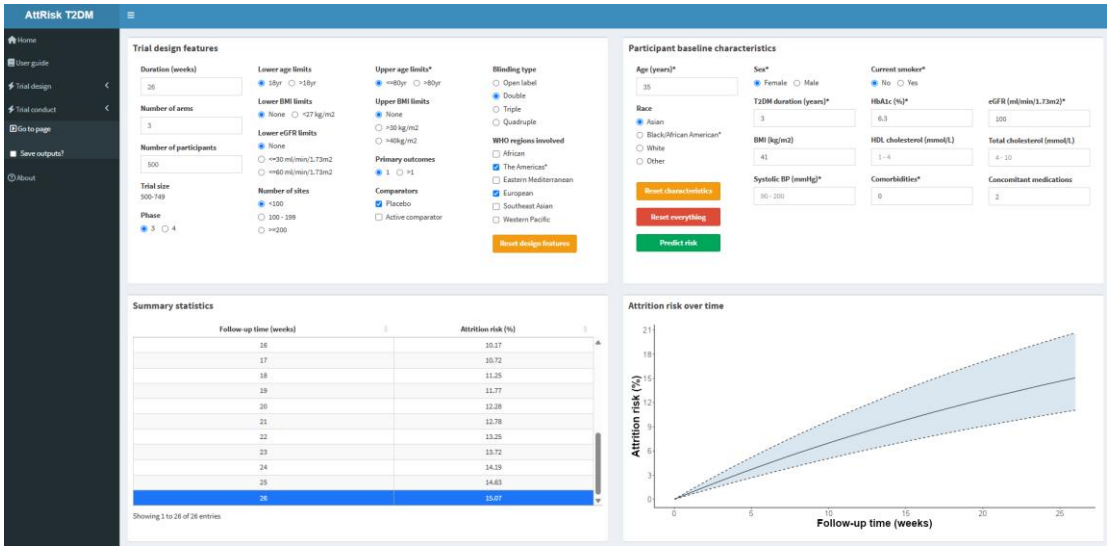


Figure 0.20 Outputs generated for participant 3 profile

Table 0.4 Attrition risk for each participant profile by the end of follow-up

Participant profile	Mean attrition risk by end of follow-up
1	22.82%
2	41.99%
3	15.07%

These comparisons could prompt a trial team to consider where additional discussion of follow-up difficulties or retention support may be useful. Support would need to reflect participants' needs and preferences where possible, and lower predicted risk would not imply that less support is needed. Characteristics can differ between individual profiles, so comparisons do not isolate the

contribution of any one characteristic or explain why a participant might not complete follow-up.

7.7 Discussion

7.7.1 Strengths and limitations

A key strength of AttRisk is the provision of a practical interface for applying the risk prediction models developed in chapter 6. Translating model outputs into a structured dashboard removes technical barriers to estimating risk, allowing users to explore predictions through clinically and operationally meaningful inputs. The separation of the application into trial design and trial conduct components also reflects the distinction between trial-level planning and participant-level risk assessment. AttRisk also includes several usability features and safeguards. Required inputs must be provided before predictions can be generated, numeric validation prevents out-of-range values from being passed to the prediction functions, and action buttons allow users to initiate prediction and clear selected inputs or the full interface. The ability to export plots and summary tables allows generated outputs to be reviewed outside the application and incorporated into discussions.

However, there were several limitations. The worked examples demonstrate functionality but are hypothetical, providing no evidence about whether intended users can complete tasks accurately, interpret uncertainty, or use outputs appropriately. The scope and performance limitations of the underlying models are discussed in chapter 6 and remain relevant to application use.

Input validation also has limitations. Checking individual numeric ranges cannot establish that a combination of trial design characteristics and participant characteristics is plausible or consistent with selected eligibility criteria. Defaults and unspecified characteristics may introduce assumptions that users could overlook. Clear presentation of these assumptions would be important when comparing profiles and interpreting outputs. Additionally, the distinction between total trial sample size and control arm denominators requires care when interpreting count summaries.

The outputs of AttRisk require careful interpretation. Risk estimates are approximate model-based predictions and should not be treated as exact forecasts of attrition over time. Where outputs are exported and used

externally, a clear statement should accompany any dissemination describing the estimates as exploratory, generated under specified assumptions, and based on models that have not yet been externally validated.

Lastly, comparisons require a user to generate and retain separate outputs, which may make it difficult to identify which inputs changed between scenarios. The application does not incorporate follow-up data or link predictions to a specific monitoring action. AttRisk therefore demonstrates a possible workflow for retention planning rather than the feasibility of effectiveness in an implemented monitoring or planning process.

7.7.2 Future extensions

Future developments of AttRisk could take several directions. The immediate priority for the interface would be formal usability testing with trialists, investigators, clinicians, and methodological researchers. Testing could have participants enter specified trial designs, compare profiles, interpret estimates, and export reproducible sets of outputs. Task completion, input errors, time taken, and interpretations of results could be assessed alongside user experiences. Testing should include users with different levels of statistical experience and assess accessibility (e.g. text size, keyboard navigation, reliance on colour). Findings could then inform interface revisions and repeat testing.

The usability of AttRisk could also be enhanced through additional functionality. Future versions could allow for simultaneous comparisons of multiple trial design scenarios or participant risk profiles, with changed inputs clearly identified. Automated report generation using RMarkdown could allow users to save rediction reports alongside input assumptions, plots, and summary tables. Both the trial design and trial conduct components could incorporate archive functionality, allowing users to revisit previous scenarios rather than recreating them manually. Clearer handling of default values, unspecified characteristics, and incompatible input combinations would further support these comparisons. Population-level predictions would also be a useful feature complimenting individual participant estimates. This could allow users to specify or import information about the distribution of participant baseline characteristics and generate a corresponding distribution of predicted attrition risks. Such a feature would be important because opposing participant-level associations identified in

chapter 6, such as age and duration of type 2 diabetes mellitus, may cancel out at the population level, influencing overall risk estimates.

Following interface refinements and adequate validation of the underlying predictions, a feasibility study could examine how AttRisk fits within trial team workflows. This could assess whether outputs meaningfully inform retention plans, the staff time required, and whether suggested priorities are acceptable to teams and participants. Evaluations would need to determine whether the use of AttRisk improves retention or monitoring efficiency, and whether there are any unintended effects.

7.8 Chapter summary

This chapter described the development and use of AttRisk, a proof-of-concept interface that incorporates risk prediction framework presented in chapter 6. The application provided two core components: a trial design component for estimating attrition risk in hypothetical trial design scenarios, and a trial conduct component for estimating participant-level attrition risk profiles within a fixed trial design. The chapter outlined the purpose, scope, structure, and functionality of AttRisk, supported by worked examples demonstrating how the application could inform retention planning and risk-based monitoring. The chapter also considered the strengths and limitations of AttRisk, and outlined future opportunities for usability testing, additional functionality, and feasibility assessments. AttRisk demonstrates the feasibility of presenting the model framework through an interactive interface. The usefulness for improving retention planning and risk-based monitoring remains to be evaluated.